Business Intelligence Analyst Assessment

Prepared for: BI Analyst Online Assessment

Dataset used: rides.csv, drivers.csv, users.csv, and support_tickets.csv

Analysis period covered by ride data: 1 July 2023 to 29 June 2024

Purpose: To diagnose operational and growth issues, define the most important BI metrics, explain the analytical methodology, and provide clear recommendations that can be used by business stakeholders.

Executive Summary:

The RideFast dataset shows a business that is generating steady ride demand, but still has several clear operational and growth risks. Across 120,000 ride requests, 81.4% were completed, which means 18.6% of requests did not become completed revenue-generating rides. The largest leakage is driver cancellation, which accounts for 11,114 ride requests, followed by user cancellation and no-show activity.

The clearest city-level operational problem is Northgate. Northgate has a completion rate of only 76.1%, compared with approximately 81.8% across the other cities combined. Its average wait time is 13.4 minutes, almost double the non-Northgate average of 7.3 minutes, and its driver cancellation rate is 13.0%, compared with 8.9% outside Northgate. This points to a local supply reliability and dispatch efficiency issue rather than a general company-wide demand problem.

Promotions are also not clearly improving conversion. Promotional rides represent 15.6% of total ride requests, but the completion rate is almost the same for promo and non-promo rides: 81.4% for promo rides versus 81.4% for non-promo rides. At the same time, completed promo rides have a lower average fare than non-promo completed rides ($28.62 versus $38.20). This suggests that discounts may be reducing revenue without producing a meaningful operational conversion uplift.

The user base also shows a serious retention challenge. Out of 35,000 registered users, 27,867 are marked as churned, equal to 79.6% of the user base. Active users have higher average wallet balance and a much higher number of promo rides than churned users, which suggests that retention and engagement need stronger lifecycle management rather than relying only on acquisition promotions.

Customer experience is another area requiring attention. The support dataset contains 22,000 tickets, of which 17.9% remain unresolved. App issues and driver behavior together represent 9,851 tickets, making them the two largest complaint categories. These issues are directly relevant to ride conversion, repeat usage, and customer trust.

Key Dataset Overview

Dataset	Records	Main use in analysis
rides.csv	120,000	Primary ride request fact table used for completion, cancellation, revenue, vehicle type, city, zone, payment, and promotion analysis.
drivers.csv	8,000	Driver supply table used for driver status, acceptance rate, cancellation rate, online hours, and driver quality analysis.
users.csv	35,000	User base table used for churn, lifetime rides, promo usage, wallet balance, and customer retention analysis.
support_tickets.csv	22,000	Customer support table used to identify complaint categories, unresolved tickets, and service quality risks.

Core Business Metrics:

Metric	Current result	Interpretation
Total ride requests	120,000	RideFast has a large operational sample for analyzing demand, cancellation, and service quality.
Completed rides	97,638 (81.4%)	Most requests are completed, but almost one in five requests still fails to become a completed trip.
Non-completed rides	22,362 (18.6%)	This is the main funnel leakage and should be monitored by city, zone, vehicle type, hour, and driver segment.
Driver cancellation rate	9.3%	Driver cancellation is the biggest failed-status category and creates lost revenue and weak user experience.
Average wait time	7.8 minutes; median 7.0;	Wait time is a leading indicator of cancellation and operational pressure.
Promo usage rate	15.6%	Promotions are present but not strongly linked to better completion performance.
Completed ride revenue	3,583,767.96	Revenue generated only from completed rides, excluding cancelled/no-show rides that have zero fare.
Average completed fare	36.70	Useful for evaluating price, discounting, and revenue quality.
User churn rate	27,867 users (79.6%)	The user base has a major retention issue that can reduce sustainable growth.
Ticket resolution rate	82.1%	Most tickets are resolved, but unresolved cases remain a service quality risk.

Diagnosis of the Main Business Problems

Based on exploratory analysis, the current growth and profitability problem is not caused by one single issue. The data points toward a combination of operational reliability, promotion efficiency, customer retention, and customer experience problems. The following five issues are the strongest business priorities.

"Problem 1: Northgate has a clear local operational failure"

Problem statement: Northgate is underperforming against the rest of the city network. It has a lower ride completion rate, higher driver cancellation rate, longer wait time, and lower user rating. This means Northgate is likely losing demand not because users do not want rides, but because the ride experience is less reliable once users request a trip.

Evidence: Northgate recorded 9,989 ride requests, but only 76.1% were completed. The average across all non-Northgate cities was about 81.8%. Northgate driver cancellation reached 13.0%, compared with 8.9% outside Northgate. Northgate also had an average wait time of 13.4 minutes, compared with 7.3 minutes outside Northgate, and user rating was lower at 3.97 versus 4.21.

Root cause hypothesis: The problem is likely related to localized supply coverage, driver positioning, dispatch efficiency, or problematic pickup zones. It is less likely to be caused only by price or promotion because the city-level operational indicators are worse even when the platform has active demand.

Recommended action: Create a Northgate recovery plan. Start with the two worst zones, NOR-Z05 and NOR-Z04, because both show very high wait times and low completion rates. Add targeted driver positioning incentives before peak periods, audit dispatch distance and driver acceptance patterns, and introduce a service-level target for pickup time in Northgate. Promotions should not be increased in this city until the operational failure is corrected, because additional demand may only increase cancellations if supply reliability remains weak.

Success metric: Increase Northgate completion rate to at least the company average, reduce average wait time below 9 minutes, reduce driver cancellation below 10%, and improve average user rating above 4.10 within one to two monthly reporting cycles.

High-risk zone	Requests	Completion rate	Non-completion rate	Avg. wait time	Promo rate
Northgate - NOR-Z05	1,967	65.5%	34.5%	22.9 min	16.1%
Northgate - NOR-Z04	1,867	65.6%	34.4%	23.0 min	16.6%
Valleyford - VAL-Z01	1,277	78.7%	21.3%	7.3 min	13.5%
Westville - WES-Z04	1,204	79.0%	21.0%	7.4 min	13.9%
Clearwater - CLE-Z01	1,232	79.1%	20.9%	7.2 min	15.6%

“Problem 2: Promotions are not generating better ride completion”

Problem statement: Promotions are being used, but the data does not show that they are improving ride completion. This creates a risk that marketing spend is subsidizing rides that may have happened anyway instead of creating incremental, sustainable demand.

Evidence: Promo rides make up 15.6% of ride requests. The completion rate for promo rides is 81.4%, while the completion rate for non-promo rides is 81.4%. The difference is operationally negligible. Completed promo rides also generated a lower average fare of 28.62, compared with 38.20 for completed non-promo rides. The average discount on promo-used rides was 9.55.

Root cause hypothesis: Current promotional targeting is likely not precise enough. Discounts may be going to already active users, price-insensitive users, or users in areas where the main barrier is availability/wait time rather than fare. In such cases, a discount does not solve the real conversion problem.

Recommended action: Stop broad discounting and move toward controlled, segmented promotion tests. Promotions should be targeted to users and routes where price sensitivity is proven: first-time users, users with abandoned requests, churn-risk users with reasonable service availability, and high-

potential city-route combinations. The company should create control groups for each campaign so the true incremental effect can be measured.

Success metric: Measure incremental completed rides per dollar of discount, promo redemption-to-completion rate, repeat ride rate after the promo, and 30-day retention of promoted users versus a control group. Promotions should continue only if they generate uplift in completed rides and retention that is higher than their cost.

Promotion segment	Ride requests	Completion rate	Average completed fare
Promo used	18,768	81.4%	28.62
No promo used	101,232	81.4%	38.20

"Problem 3: High user churn shows weak long-term retention"

Problem statement: RideFast has a large registered user base, but the churn flag indicates that most users have not taken a ride recently. This is a major strategic issue because sustainable growth depends on repeat customers, not only new acquisition.

Evidence: 27,867 out of 35,000 registered users are marked as churned, equal to 79.6% of the user base. Active users have an average of 34.1 lifetime rides and $69.55 wallet balance, compared with 30.1 rides and $39.89 wallet balance for churned users. Active users also have much higher average promo-ride history, which suggests that engagement mechanics and incentives can influence repeat usage if used carefully.

Root cause hypothesis: Users may be dropping out after poor ride experiences, high wait times, unresolved service issues, or lack of relevant engagement after the first few rides. Churn can also happen when promotions are used only as one-time acquisition tools instead of being connected to a lifecycle journey.

Recommended action: Build a retention-focused lifecycle programmed. Users should be segmented into new users, active repeat users, at-risk users, and churned users. Each segment should receive different treatment: onboarding education for new users, loyalty benefits for repeat users, service recovery offers for users after cancelled or poor-rated trips, and win-back campaigns for churned users in cities where service quality is strong enough to support return demand.

Success metric: Reduce churn rate, increase 30-day repeat ride rate, increase rides per active user, and improve retention after a user receives a service recovery or win-back intervention. Track retention by city and by previous ride experience rather than only at total-company level.

User group	Users	Avg. lifetime rides	Avg. promo rides	Avg. wallet balance
Active	7,133	34.1	11.2	69.55
Churned	27,867	30.1	3.3	39.89

"Problem 4: Customer support issues are large enough to affect trust and repeat usage"

Problem statement: Support tickets are concentrated in app issues, driver behavior, and fare disputes. These are not minor back-office issues; they directly affect trust, perceived safety, booking confidence, and repeat usage.

Evidence: The dataset contains 22,000 support tickets. The largest categories are app issues (4,983 tickets), driver behavior (4,868 tickets), and fare disputes (4,642 tickets). Overall, 17.9% of tickets are unresolved. Driver behavior tickets take an average of 5.9 hours to resolve, and safety tickets take 5.3 hours on average.

Root cause hypothesis: App and driver experience problems may be creating friction after booking or after trip completion. If these issues are not resolved quickly, customers may become less willing to request rides again, especially for higher-value intercity or longer-distance trips where trust matters more.

Recommended action: Introduce a customer experience improvement programmed. App issue tickets should be mapped to specific screens and release versions. Driver behavior and safety tickets should trigger driver coaching, monitoring, and escalation rules. Fare disputes should be analyzed against surge, promo discount, payment method, and final fare to identify confusion or incorrect expectation-setting.

Success metric: Reduce app issue ticket rate per 1,000 rides, reduce unresolved ticket rate, reduce average resolution time for safety and driver behavior issues, and increase repeat ride rate after support cases are resolved.

Ticket category	Tickets	Unresolved rate	Avg. resolution time
App Issue	4,983	18.0%	3.8 hrs
Driver Behavior	4,868	18.0%	5.9 hrs
Fare Dispute	4,642	18.5%	3.9 hrs
Lost Item	4,212	16.6%	4.0 hrs
Safety	3,295	18.1%	5.3 hrs

Problem 5: Cancellation leakage is reducing revenue and hiding behind acceptable overall averages

Problem statement: The company-wide completion rate appears acceptable at first glance, but the non-completed ride volume is large enough to represent a serious revenue and customer-experience leakage. The issue becomes more important when viewed by city, zone, status, and wait time.

Evidence: 22,362 ride requests did not become completed rides. Driver cancellations alone account for 11,114 requests, user cancellations account for 8,198, and no-shows account for 3,050. Average wait time is higher for failed outcomes: driver-cancelled rides average 10.3 minutes, user-cancelled rides average 8.6 minutes, no-shows average 9.2 minutes, while completed rides average 7.4 minutes.

Root cause hypothesis: A material portion of ride failure is likely driven by waiting-time pressure, driver reliability, and mismatch between requested demand and nearby available supply. The issue is not evenly distributed; it is especially visible in Northgate and in certain pickup zones.

Recommended action: Treat cancellation as a core growth metric, not only an operational metric. Build a cancellation dashboard by city, zone, hour, vehicle type, driver cohort, and wait-time bucket. Use this dashboard to trigger operational action, including driver incentives where dispatch coverage is low, driver penalties or coaching for repeated accepted-then-cancelled rides, and improved ETA communication for users.

Success metric: Reduce total non-completion rate, reduce driver cancellation rate, lower average wait time for failed requests, and increase completed ride revenue without increasing discount cost.

Ride status	Requests	Share of all requests	Average wait time
Completed	97,638	81.4%	7.4 min
Cancelled Driver	11,114	9.3%	10.3 min
Cancelled User	8,198	6.8%	8.6 min
No Show	3,050	2.5%	9.2 min

Evaluation of Current Targeting Methods

The current targeting approach should not be judged only by redemption rate, because redemption can look high even when the business is paying discounts to users who would have taken rides anyway. The better question is whether targeting creates incremental completed rides, repeat usage, and profitable retention.

Based on the available data, the current promotion approach appears weak because promo rides and non-promo rides have almost identical completion rates. This does not prove that all promotions are ineffective, but it does show that the present data does not support broad discounting as a reliable growth lever. Targeting should therefore be evaluated using controlled experiments rather than campaign totals only.

Targeting question	How to evaluate it	Required data points
Are promotions reaching the right users?	Compare promoted users with a matched non-promoted control group and measure incremental completed rides and repeat rides.	User ID, campaign exposure, promo use, ride status, ride date, fare, discount, city, previous rides, churn flag.
Are city and zone campaigns effective?	Measure uplift by city, pickup zone, and peak period, while controlling for supply availability and wait time.	City, pickup zone, request time, driver availability proxy, wait time, completion status, promo used.
Are cross-sell campaigns working?	Track conversion from other vertical users into first intercity/ride request and second ride within 30 days.	User segment, campaign exposure, first ride date, repeat ride date, completed rides, cancellation status.
Are new product features improving booking?	Run A/B tests or pre/post analysis on booking completion, cancellation, time-to-book, and support tickets.	Feature exposure flag, booking step logs, request time, status, app issue tickets, user cohort.

Analytical Methodologies

The analysis should combine descriptive BI reporting, diagnostic segmentation, and controlled experimentation. The following methods are recommended for a complete BI assessment.

Method	Purpose	Application in this case
Exploratory data analysis	Understand data quality, distributions, trends, and outliers before dashboarding.	Validate row counts, date ranges, missing values, ride status distribution, fare values, wait time, and ticket categories.
Funnel analysis	Measure how many requests move from request to pickup to completion.	Track request volume, completed rides, driver cancellation, user cancellation, and no-show rate by city, zone, hour, vehicle type, and promo usage.
Cohort analysis	Understand retention over time.	Group users by signup month or first ride month and measure repeat ride rate after 7, 30, and 60 days.
Promotion incrementality analysis	Determine whether discounts create new completed rides or only	Compare treated users with matched control users using pre-campaign behaviour, city, ride frequency, and churn risk.

	subsidise existing demand.	
A/B testing	Evaluate product features or new campaign rules with causal confidence.	Randomly expose users to a new booking feature or promotion design and compare booking completion, cancellation, repeat rides, support tickets, and revenue.
Driver supply diagnostics	Identify operational causes of cancellation and long wait times.	Analyse driver acceptance rate, cancellation rate, online hours, status, city, vehicle type, and relationship with ride outcomes.
Root-cause segmentation	Move from total-company averages to actionable problems.	Break down Northgate and zone-level issues, especially NOR-Z04 and NOR-Z05, rather than treating all cities equally.
Support ticket analysis	Understand customer pain points and quality risks.	Track ticket category, severity, resolution status, resolution time, and repeat usage after support resolution.

Strategic Recommendations: Stop, Start, Continue

Decision	Recommendation	Reason
Stop	Stop relying on broad discounting as the main growth tool.	Promo and non-promo completion rates are almost identical, while promo rides have lower average completed fare.
Stop	Stop reading company-wide averages without city and zone breakdowns.	Northgate and specific zones show much worse performance than the network average.
Start	Start a Northgate and high-risk-zone operational recovery programme.	Long wait times and higher driver cancellations indicate local operational friction.
Start	Start controlled promotion experiments with holdout groups.	The company needs to know whether promotions are incremental, not just whether they are redeemed.
Start	Start user lifecycle and churn-risk management.	The churn rate is very high and growth will not be sustainable without better repeat usage.
Start	Start tracking support experience as a growth metric.	App issues, driver behaviour, and fare disputes can reduce trust and repeat rides.
Continue	Continue monitoring supply-side metrics such as driver acceptance, cancellation, and online hours.	Supply reliability has a direct relationship with wait time, cancellations, and user experience.
Continue	Continue product feature improvements, but evaluate them with experiments.	New features should be judged by measurable improvement in booking completion, cancellation, and support outcomes.

Priority Action Plan

Priority	Action	Owner	Time frame	Measurement
1	Northgate operational recovery: focus on NOR-Z04 and NOR-Z05, driver positioning, peak coverage, dispatch audit, and cancellation monitoring.	Operations and Driver Supply	First 2-4 weeks	Completion rate, average wait time, driver cancellation, user rating.
2	Promotion redesign: replace broad discounts with targeted A/B tests and control groups.	Growth and BI	First 4 weeks	Incremental completed rides per discount dollar, repeat ride rate, net revenue.
3	Churn prevention programme: segment users by lifecycle stage and trigger retention actions after poor experiences.	CRM and Product	4-8 weeks	30-day repeat ride rate, churn rate, active users, rides per active user.
4	Customer experience improvement: reduce app issues, fare disputes, and driver behaviour complaints.	Product, Support, Driver Quality	4-8 weeks	Ticket rate, unresolved rate, resolution time, repeat ride after ticket.
5	BI performance review cadence: weekly operations dashboard and monthly strategic growth review.	BI and Leadership	Ongoing	Progress against target metrics and actions completed.

Long-Term Sustainable Growth Strategy

For long-term competitive advantage, RideFast should move from promotion-led growth to reliability-led growth. In ride-hailing, customers may try the service because of a discount, but they continue using it when the ride is dependable, drivers arrive on time, prices are transparent, and issues are resolved quickly. Therefore, the company should treat operational reliability and customer trust as growth assets.

The long-term plan should combine three layers. First, build strong operational intelligence by monitoring completion, wait time, cancellation, and driver behaviour at a granular city-zone-hour level. Second, improve customer lifecycle management by identifying the right users for onboarding, retention, service recovery, and win-back campaigns. Third, use controlled experiments for product changes and promotions so the company invests only in strategies that create measurable incremental value.

The most important competitive edge will come from using BI not only to describe what happened, but to decide what to do next. The dashboard should therefore be connected to action rules: when a zone crosses a wait-time threshold, driver incentives or dispatch changes should be triggered; when a user has a poor ride or unresolved ticket, a recovery flow should begin; when a promotion does not increase incremental completed rides, it should be paused or redesigned.

Conclusion

RideFast has a usable data foundation and a meaningful ride volume, but the analysis shows clear opportunities to improve performance. The strongest issues are Northgate underperformance, weak promotion incrementality, high user churn, customer support friction, and cancellation leakage. These problems should be addressed through a combination of targeted operational fixes, better promotion testing, retention management, and customer experience improvement.

The recommended strategy is to stop using broad discounts as the main answer, start correcting city and zone-level service reliability, and continue building dashboard-driven decision-making. By linking BI metrics to operational action, RideFast can improve completion rates, protect revenue, retain more users, and create a more sustainable competitive position.

Assessment Link:

Looker Studio (Google Data Studio) Link